# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

Project ID: R26-IM-001

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

| Comment 1 (IT22641410) | How we addressed them |
| --- | --- |
| Whats the logic candidate use to reduce cognitive workload. The logic used to compare control group and experiment group is incorrect. | Cognitive Workload Logic Clarified: Applied Multimedia Learning principles, specifically Split-Attention Reduction.<br><br>The system now uses spatial integration (locking AR elements directly to the physical page) and sequential, guided prompts rather than presenting all information simultaneously, directly minimizing visual clutter and extraneous mental effort for Grade 1 learners.<br><br>Experimental Design Corrected: Redefined the A/B Testing variables to ensure valid comparison.<br><br>Control Group: Evaluates a standard, static AR interface (baseline).<br><br>Experimental Group: Evaluates the proposed "Hybrid Interaction Framework" (Adaptive UI + AR-to-2D Transition + Sequential Prompts).<br><br>Metrics explicitly isolate cognitive load using facilitator-assisted simplified NASA-TLX questions and task-completion times. |

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

| Comment 2 (IT22179258) | How we addressed them |
| --- | --- |
| This component overlaps with IT22641410 candidates component. | We clearly separated responsibilities:<br>HIF owns the interaction and AR placement layer:-raw touch input, page scanning, anchor tracking, UI layout, and cognitive-load control<br>while AVM owns only the 3D plant visual layer:-<br>plant state animations, color changes, glow/particle effects, audio cues, and growth-stage visuals.<br>AVM now responds only to processed events and placement data provided by HIF, rather than handling interaction or AR tracking directly. |

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

| Comment 3 (IT22237804) | How we addressed them |
|---|---|
| Candidate needs to explain what types of gamified elements that would be implemented in the proposed component. | Stewardship-Based Progression – Story progress depends on caring for the plant, not collecting points.<br>Consequence-Based Story Branching – Learner choices change the plant condition and the next story outcome.<br>Redemption / Recovery Mechanic – Children can correct harmful or neglectful actions through later care actions.<br>Narrative Feedback and Plant-State Responses – the system provides immediate feedback through story changes linked to plant health and recovery. |

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

| Comment 4 (IT22626974) | How we addressed them |
| --- | --- |
| Current AR model already maps realword environments to simulated environment right? Discuss with supervisor regarding this component. | Existing AR systems mainly use environment mapping for tracking, while our component uses IoT data and context-aware simulation to dynamically control virtual plant growth based on real-world environmental conditions and user interaction. |

## Research Problem

"How can primary learners studying the Cambridge Grade 1 Science module in Sri Lanka be better supported in understanding plant growth and environmental changes that are difficult to learn through traditional text-based teaching methods?"

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

| Impact on Users | Impact on Society |
|---|---|
| Weak understanding of plant growth and plant care | Poor environmental awareness from an early age |
| Low engagement with science learning | Less empathy toward living things and nature |
| Difficulty visualizing biological processes from static content | Missed opportunity to build responsible future citizens |
| Reduced curiosity and classroom participation | Lower quality of innovative and inclusive education |

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

## Users and Stakeholders

### Persona 1 : Primary Learner

**Age:** 6 – 8 years

**Role:** Primary User



**Needs:**

- Fun and interactive visual learning
- Simple and age-appropriate content
- Easy understanding of plant growth and changes

**Wants:**

- To understand how plants grow, react, and survive
- To feel engaged and curious during the lesson

### Persona 2 : Teacher

**Role:** Secondary Stakeholder



**Needs:**

- Engaging teaching support tools
- Clear and interactive lesson delivery
- A system that helps explain plant concepts easily

**Wants:**

- To improve student attention and participation
- To make science lessons more effective
- To use technology that supports classroom teaching

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

## Users and Stakeholders

### Persona 3 : Parent

**Role:** Secondary Stakeholder



**Needs:**

- Safe, meaningful, and age-appropriate learning
- A tool that supports the child's understanding both at school and at home

**Wants:**

- To support the child's learning progress
- To encourage interest in science and nature
- To ensure the learning experience is both useful and enjoyable

### Persona 4 : School

**Role:** Secondary Stakeholder



**Needs:**

- Accessible and innovative learning tools
- Technology that improves classroom engagement
- Educational solutions suitable for young learners

**Wants:**

- To enhance the quality of teaching and learning
- To adopt modern and interactive classroom methods
- To support better science education outcomes

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

## Design Excellence

- Child-centered AR storybook designed for Grade 1 plant learning
- Combines physical storybook + AR interaction + environmental response + narrative feedback
- Makes abstract plant concepts easier through:
  - interactive 3D plant visualization
  - story-based learning flow
  - real-world context awareness
  - emotionally meaningful feedback
- Designed to improve:
  - understanding of plant growth
  - engagement and curiosity
  - environmental care and stewardship

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

## Why the Overall Solution is Strong

- Each component solves a distinct learning challenge:
  - **C01** -HIF: child-friendly AR interaction
  - **C02** -AVM: understandable plant health and response visuals
  - **C03** -GNE: meaningful story progression and responsibility
  - **C04** -CASE: realistic environmental simulation
- Together, they create a complete, immersive, and curriculum-aligned learning experience.





# IT22641410 | Wijayarathne K.N.

## Hybrid Interaction Framework for an AR Storybook

# Hybrid Interaction Framework for an AR Storybook

How can a **child-centered hybrid interaction framework** mitigate **touch inaccuracy, AR instability, the "Gorilla Arm Effect," and cognitive load** to enhance **the usability and learning effectiveness** of handheld AR storybooks for Grade 1 learners?

# Hybrid Interaction Framework for an AR Storybook

## Solution Approach & User Requirements

### HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM

**The Core Challenge**: Grade 1 motor constraints lead to rapid physical fatigue (Gorilla Arm Effect) and touch inaccuracy during continuous AR tracking.

- **Solution 1:** Seamless AR-to-2D Transition
- **Mechanism**: System logic tracks holding time and transitions the 3D AR view to a stable 2D animation at exactly 45 seconds.
- **Impact:** Acts as an ergonomic "rest cycle," allowing the child to place the tablet flat, directly mitigating shoulder strain.

- **Solution 2:** Adaptive UI Zones
- **Mechanism:** The interface dynamically calculates grip position and shifts actionable buttons into localized "Thumb-zones."
- **Impact:** Accommodates fine-motor variability and prevents the physical stretch-and-aim fatigue common in standard mobile UIs.

# Hybrid Interaction Framework for an AR Storybook

## Solution Approach & User Requirements

**User Requirements addressed by the solution:**

**FR3 (Dynamic UI Resizing):** The system dynamically resizes and adjusts interactive controls in response to the learner's touch variability and device shake.

**FR4 & FR5 (Stable Anchoring & Occlusion):** Utilizes page-based physical anchors to ensure 3D plant overlays remain locked to the storybook, even when hands partially cover the page.

**FR6 (Ergonomic Interaction Flow):** Hard-codes interaction cycles into short, 30-to-40-second bursts to actively prevent prolonged arm elevation.

**FR7 (Split-Attention Reduction):** Deploys guided digital signposts that keep the learner's visual focus centralized.

# Hybrid Interaction Framework for an AR Storybook

## Contribution & Prototype Feedback

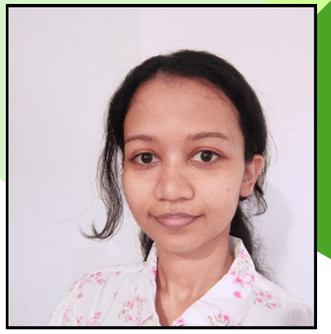
### Design Excellence / Contribution:

- **Fatigue-aware Interaction Framework:** A novel, child-centered architectural module that actively minimizes physical fatigue and cognitive overload by dynamically resizing UI elements in real-time.

- **Automated Ergonomic Interventions:** The system does not rely on the child to recognize fatigue; it automatically intervenes (via the AR-to-2D transition or gamified rest prompts) to maintain physical comfort.

### User Feedback on Prototype:

- **Subjective Comfort Measurement:** Utilizing the child-friendly Borg RPE Scale (Emoji scale ranging from a Happy to a Tired face) to successfully gather perceived exertion feedback from 6-year-olds.
- **Objective Fatigue Tracking:** Logged observational metrics, specifically "Arm Drop Count" and "Hand Switching," to physically quantify muscle fatigue during standard use.
- **A/B Testing Validation:** Compared traditional "Continuous AR" sessions against the proposed "Hybrid Transition" strategy to calculate and prove the exact percentage reduction in physical fatigue.



# IT22179258 | Palliyagurguge.S.N

## Affective Visualization Model

# Affective Visualization Model for AR Story Book

**“How can an AR storybook clearly and emotionally communicate plant health, stress, and recovery states to young children in a way that improves understanding and encourages empathy toward plants?”**

# Affective Visualization Model for AR Story Book

## How the Solution Addresses the Sub-Problem

The proposed **Affective Visualization Model** transforms plant condition changes into clear, emotionally meaningful AR feedback that young children can understand without relying only on text or verbal explanation.

The prototype represents key plant states through carefully designed visual responses:

- A **healthy plant** is shown through a bright, lively appearance enhanced with subtle sparkle effects.
- **Environmental responsiveness** is communicated by leaves gently turning toward the direction of sunlight.
- A **stressed plant** is represented through paused growth, reduced color vibrancy, and a visibly wilted posture.
- During **recovery**, the plant gradually regains its healthy color and posture, and its growth continues.

# Affective Visualization Model for AR Story Book

## User Requirements Addressed

The solution is designed around the perceptual and learning needs of young children by addressing four key user requirements:

- **Clarity of interpretation:** Children can distinguish healthy, stressed, and recovering plant states through direct visual cues.
- **Cause-and-effect understanding:** The system helps learners connect care actions and environmental conditions with visible plant responses.
- **Child-friendly feedback:** Plant state changes are communicated through simple, age-appropriate animations, effects, and color cues.
- **Emotional engagement:** The visual responses encourage children to feel concern for the plant and recognize the importance of responsible plant care.

# Affective Visualization Model for AR Story Book

## Design Excellence / Contribution

### Design Excellence / Contribution

- Introduces a state-based affective visualization model for plant learning
- Goes beyond static AR plant models and growth-stage display
- Combines: animation,posture changes,color transitions,visual effects
- Shows not only how a plant grows, but how it responds to care or neglect
- Supports emotionally meaningful, story-linked interaction for young learners.

# Affective Visualization Model for AR Story Book

## User Feedback Collection & Validation

**Visual State Recognition Task:**

- Children will identify whether the AR plant is healthy, stressed, or recovering using simple child-friendly responses. Accuracy will measure how clearly the affective visual cues communicate plant conditions.

**Observation-Based Engagement Measurement:**

- Facilitators will record visible indicators such as attention, curiosity, emotional reaction, and responsiveness when plant color, posture, and recovery effects appear.

**Learning Effectiveness Validation:**

- A control group using traditional teaching will be compared with an experimental group using the AR prototype to evaluate improvement in understanding plant health, stress, and recovery.

**Teacher / Facilitator Feedback:**

- Structured feedback will be collected on the clarity, age-appropriateness, and educational usefulness of the plant animations, color changes, and visual effects.



# IT22237804 | Amarawardhane A.S.

## Gamified Narrative Engine for a Handheld AR Storybook

# Gamified Narrative Engine for a Handheld AR Storybook

Research Objectives

**“How to Design a Gamified Narrative Engine That Builds Children’s Environmental Responsibility Through AR Storytelling?”**

# Gamified Narrative Engine for a Handheld AR Storybook

## Solution Approach & User Requirements

### HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM

**The Core Challenge:** Traditional AR educational stories are often fixed, short-term, and reward-focused, so they do not effectively build long-term responsibility, care, or emotional engagement in young learners.

- **Solution 1: Stewardship-Based Persistence**
- **Mechanism:** Plant condition and story progression depend on the child's care actions over time, rather than points or badges.
- **Impact:** Encourages responsibility by making continuous plant care meaningful to the story.

- **Solution 2: Non-Linear Story Branching**
- **Mechanism:** The story changes according to plant condition, learner choices, and previous care behavior.
- **Impact:** Creates a responsive narrative where children experience the consequences of their actions.

# Gamified Narrative Engine for a Handheld AR Storybook

## Solution Approach & User Requirements

**User Requirements addressed by the solution:**

**FR2 & FR3 (Adaptive Narrative Flow):** Generates rule-based story events and adapts progression according to learner choices, plant condition, and care behaviour.

**FR4 (Persistent Story State):** Maintains plant and narrative continuity across sessions, so the story reflects ongoing care over time.

**FR5 & FR8 (Child-Friendly Dialogue & Feedback):** Provides age-appropriate prompts and clear story feedback that help children understand how their actions affect the plant.

**FR6 & FR7 (Meaningful Gamification & Redemption):** Uses care-based story progress and recovery opportunities instead of points, while allowing learners to correct earlier mistakes through positive actions.

# Gamified Narrative Engine for a Handheld AR Storybook

## Contribution & Prototype Feedback

**Design Excellence / Contribution:**

- **Stewardship-Based Narrative Progression:** A child-centred gamification approach where story progress depends on plant care, rather than points or badges.
- **Adaptive Story Consequence Logic:** Rule-based branching changes dialogue, plant condition, and story direction according to learner choices.

**User Feedback on Prototype:**

- **Clear Cause-and-Effect Understanding:** Learners could recognize that their care actions influence the plant's condition and story outcome.
- **Engaging Care-Based Interaction:** Sunlight, watering, and recovery-related story responses made the prototype more meaningful and interactive.
- **Improvement Direction:** Feedback suggested expanding story branches and refining dialogue transitions for a smoother narrative experience.

# IT22626974 | Dissanayake D.M.D.D.



## Context-Aware Simulation Engine for a Handheld AR Storybook

# Context-Aware Simulation Engine for a Handheld AR Storybook

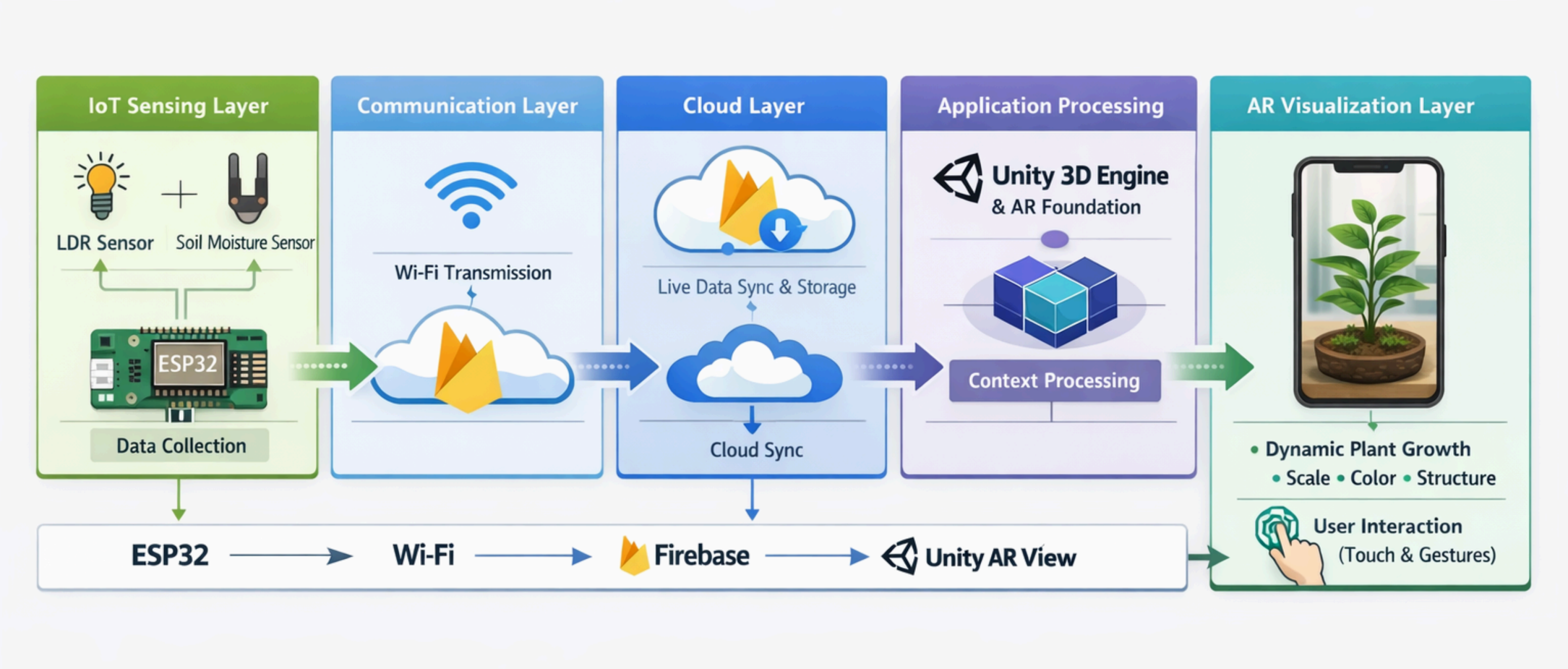
How to design **“a Context-Aware Simulation Engine respond to real-world environmental conditions and user interactions in real time ? “**

# Context-Aware Simulation Engine for a Handheld AR Storybook

## Core Problem

Current AR systems mainly use environmental data for visual effects and tracking, but **do not integrate real-world conditions into simulation behavior**. As a result, virtual plant models remain **static and fail to respond dynamically** to changes in light, user interaction, limiting realism, interactivity, and learning effectiveness.

# Context-Aware Simulation Engine for a Handheld AR Storybook



# Context-Aware Simulation Engine for a Handheld AR Storybook

## Hardware & IoT Solution:

**Architecture:**

Custom ESP32-based IoT environmental monitoring node integrating:

- DHT22 Temperature & Humidity Sensor
- LDR Light Intensity Sensor
- SI1145 UV/IR Sensor
- Capacitive Soil Moisture Sensor

for real-time environmental sensing and adaptive AR plant growth simulation.

**Communication:**

Engineered a real-time Firebase cloud synchronization pipeline between:

- ESP32 IoT node
- Firebase Realtime Database
- Unity AR application

enabling live environmental data transfer and dynamic AR updates.

# Context-Aware Simulation Engine for a Handheld AR Storybook

## Key Metrics Achieved

- **Real-Time Synchronization:**

Achieved low-latency environmental data updates between ESP32 and Unity AR simulation through Firebase Realtime Database synchronization.

- **Environmental Responsiveness:**

Successfully mapped real-world environmental changes such as:

  - sunlight intensity , UV exposure , humidity , temperature , soil moisture

into dynamic AR plant growth behavior in real time.

- **Adaptive Interaction:**

Implemented flashlight-based artificial sunlight simulation allowing PWM LED flashlight interaction to influence virtual photosynthesis and plant growth.

# Context-Aware Simulation Engine for a Handheld AR Storybook

## Design Excellence:

- Context-Aware Environmental Intelligence (IoT sensor data and AR interaction)

- Cyber-Physical Integration (dynamically control virtual plant health, growth rate, and animation states.)

- Flashlight-Based Photosynthesis Simulation (PWM LED)

- Spatial Context Awareness (Integrated Unity AR Foundation environment mapping and surface recognition to analyze user surroundings and adjust plant growth conditions dynamically.)

# Context-Aware Simulation Engine for a Handheld AR Storybook

## User Feedback on Prototype:

- The real-time environmental interaction more engaging than static AR plant simulations.
- The flashlight-based artificial sunlight feature was identified as intuitive and interactive.
- Users responded positively to dynamic plant behavior changes based on temperature, light, and humidity conditions.
- The IoT-integrated AR experience improved realism and environmental awareness.
- Real-time sensor-driven feedback helped users better understand the relationship between environmental conditions and plant growth.

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship - Commercialization

## Commercialization / Sustainability of the Solution

### Customer Persona

- **Buyers:** International schools and school decision-makers
- **Users:** Grade 1 teachers and primary learners
- **Secondary audience:** Parents supporting guided home learning

### Market Size

- **Initial focus:** Sri Lankan international schools using Cambridge Primary Science
- **Future expansion:** private schools, home-learning kits, and additional science modules

# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship - Commercialization

## Commercialization / Sustainability of the Solution

Unique Value

- A curriculum-aligned AR storybook that combines:
  - physical book learning
  - interactive AR
  - emotional plant-state visualization
  - adaptive story and real-world environmental response

Cost Recovery

- Estimated cost: LKR 54,000
- Recovered through:
  - school subscriptions
  - class/device licensing
  - storybook + app bundles
  - future content expansion packs